

Thermal and hydraulic analysis of slotted plate fins heat sinks using numerical and experimental techniques

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HIGHLIGHTS

- Performance analysis of slotted plate-fin heat sinks.
- Hydraulic and thermal analysis of the heat sinks.
- Mass reduction of the plate fins.
- Analytical model validation with experiments.
- Utilization of 3D printing techniques for the development of various elements.

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ABSTRACT

A heatsink's thermal and hydraulic performance can be optimized by making several modifications in the geometry. Modifying the heatsinks usually increases the weight by adding material. This study focuses on enhancing heat transfer by the removal of material in the form of slots. Fabrication of slots is one of the advantageous geometric modifications in the plate fins. This study also focuses on optimizing the slots for heat transfer augmentation without compromising the pressure drop. In this study, the authors performed Computational Fluid Dynamics (CFD) analysis of various geometric iterations of the plate fins. The heat sinks with comparatively better thermal performance are pointed out (S8 and S9). On average, the Nusselt number of the heat sink S9 is 28.4% higher than the plain heatsink P at various Reynold's number values. The introduction of slots drops the pressure across the heat sinks with an additional average of 9.3% pressure drop across the heatsink S9 compared to the heatsink P at various Re number values. The authors performed an experimental analysis to validate the numerical analysis. The maximum difference between the observed and simulated base temperature values is 2.6%, 2.9%, and 3% for the heat sinks P, S8, and S9, respectively.

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Acronyms

CFD	Computational Fluid Dynamics
SLM	Selective Laser Melting
RANS	Reynolds-Averaged Navier-Stokes
SST	Shear Stress Transport
SIMPLE	Semi Implicit Method for Pressure Linked Equations
DC	Direct Current
AC	Alternating Current
FFF	Fused Filament Fabrication

Symbols

H	Height of fins
L	Length of fins
W	Width of the heat sink
s	Distance between consecutive fins
t	Thickness of fins
t _{base}	Thickness of the heat sink base
$Re = \frac{\rho_{air} U_{air} D_h}{\mu_{air}}$	Reynold's number
ρ_{air}	Density of air
U	Speed of air
$D_h = \frac{4A_c}{P}$	The hydraulic diameter of the test section
μ	Dynamic viscosity of air
A _c	The cross-sectional area of the test section
P	The perimeter of the test section
k _s	Conduction coefficient
T	Temperature
C _p	Specific heat at constant pressure
Pr	Prandtl number
$\dot{Q}$	Heat transfer rate
P _k	Production limiter of turbulence
S	Invariant strain rate
F	Blending function
Δx	Insulation thickness
h _p	Convection coefficient calculated using projected area of the heat sink
$Nu_p = \frac{h_p L}{k_{air}}$	Nusselt number calculated using projected area of the heat sink
ΔP	Pressure-drop
m	mass
V	Volume

1. Introduction

The caloric theory suggested by Lavoisier indicated that heat is a mass-less fluid called caloric. Mareš et al. described that the temperature increases or decreases when a body possesses heat or the mass-less fluid [1]. James P. Joule, 1843 further investigated this caloric theory, and the experiment indicated that heat is transmitted from one state to another due to temperature differences [2]. Heat dissipation is considered a crucial factor in most engineering applications. Heat accumulation may result in an efficiency drop as well. In the application of Internal Combustion Engines (ICE), about 25% of the generated power gets converted into valuable mechanical energy. In contrast, more than 30% of the generated heat is dissipated through engine cylinder walls [3].

In other applications like electronics, the elements of an electronic circuit like transistors, resistors, and inductors generate enough heat to affect the performance and durability of the apparatus. So, to rectify this issue, fins are used in electronic circuits to control the excess heating. Fins are extended surfaces with a larger exposed area to enhance the heat transfer by convection [4].

Researchers observed the effects of various configurations of the delta winglets on the heat transfer and pressure drop [5]. Previous studies in the applications of fins for better heat dissipation indicated that various alterations were done in the fin structure like depressions, voids, slots, holes, and channels. These alterations in the geometric design provided an additional surface area for more effective heat transfer and decreased the build material of the fins.

The study by Adhikari et al. characterized the optimum heat transfer to determine a comprehensively optimized fin design. The application of the dynamic-Q algorithm found that design parameters for the fin design, including height, length, and layout, greatly

influence the fluid stream through the fin channels to improve the heat dissipation rate from the fins [6]. In a numerical study by Alfellag et al. the authors conducted a parametric study by altering the height and angle of the inclined slot and the pin position for the inclined slotted plate-fin to optimize the hydrothermal performance. The simulation results concluded that slots and pins complement each other for optimum hydrothermal performance. They reported a higher Nusselt number for the pin position closer to the leading edge of the slot [7].

Jeon and Byon studied the natural convection phenomenon for plate-fin heat sinks having fins with double-height profiles. Different variables were numerically studied, such as dual-height configuration, channel length, and fin spacing. They observed that thermal performance per unit mass increased when the height of the second fin was decreased. They also concluded that the dual-height heat fin concept was applicable above a certain threshold height of the prominent fin, and the thermal performance started degrading after that [8].

Feng et al. studied the application of double circular pin-fins in an improved heat transfer performance analysis. The pin-fin spacing and height are significant factors for enhancing the heat transfer rate and reducing pressure drop. Simulations were conducted for a range of 133–530 Reynolds number. The optimized parameters of 0.08 mm height, the diameter of 12 mm, and spacing of 0.55 mm for a Reynolds number of 530 enhanced heat transfer by 42% compared to smooth microchannels [9]. Another investigation by Ozdilli and Sevik introduced models of trapezoidal curved-shaped plate-fins heat sinks, one with a rectangular channel and the other with a rectangular corner channel. Significant improvement in convective heat transfer was achieved for the models compared to the standard plate-fin heat sinks. Moreover, improved thermal resistance was also observed in natural convection [10].

Al-Sallami et al. made a numerical comparison of different design configurations like notches and slots with various circular holes in plate-fin heat sinks to investigate heat transfer with pressure drop. Results from the conjugate model showed that all modifications improved heat transfer and pressure drop at some level. However, fins with slotted perforations have proved to be the most efficient design for enhancing heat transfer rate with reducing pressure drop simultaneously [11]. The application of fins in heat exchangers induces an increase in heat transfer rate; however, it could cause pressure loss in the system, demanding some extra pumping energy. Heat sink application can improve the heat dissipation rate, but this can cause a pressure drop. Numerous geometric features were integrated into the heat sink design structure [12].

Moreover, fins increase the system weight by adding additional material, escalating the overall cost. The heat sink needs to have less weight in electronic applications while being compact and practical. This objective can be achieved by introducing perforations and interruptions in the fin design. The heat transfer rate can be improved by applying perforations as they have extra surface area for convective heat transfer and could aid in reducing pressure loss and heat sink weight [13,14]. A study done by Wang et al. investigated a dual-layered microchannel design that yielded 45.3–48.5% less reduction in pumping power for the same Reynolds number. Moreover, it was also concluded that cooling performance depended mainly on a more significant channel aspect ratio to achieve lower thermal resistance [15]. Alfellag et al. introduced microchannels in a plate-finned oval pins sink for slotted and non-slotted configurations under 100 to 1200 range of Reynolds number. Their analysis contains specific observations against different governing parameters [16]. Narato et al. studied heat transfer and flow characteristics for inclined cylindrical pins employed at various angles ranging from 30 to 150°. The numerical simulation concluded that only pin arrays at 120° and 135° successfully enhanced the heat transfer compared to the standard vertical pin. Moreover, pin inclination angles of 120°, 135° and 150° indicated a reduction in pressure drop and enhanced thermal performance [17].

Sahin and Demir reported their findings on the improvement of heat transfer and the subsequent drop of pressure around a flat surface having square cross-sectional pierced pin fins in a channel with a rectangular cross-section. Their investigation indicated that the application of the square pin fins could improve the heat transfer rate. They concluded that the best results were observed at the Re (Reynold number) 42000, 50 mm fin height, and distance of 51 mm among fins [18].

Dorignac et al. described in their study the effects of the convective heat transfer on a pierced plate. They reported the heat transfer due to the airflow on the perforations of smaller and larger diameters. The perforations were vertical with the walls. In their study, an experimental relation was suggested for heat transfer at the windward surface of the pierced plate [19]. More investigations in this area verified that the fin tip temperature was lower with perforations than without perforations. Additionally, the material used in the fabrication was reduced, and the pressure decrease across the heat sink was reduced too [20–22].

The fabrication of heat sinks is another topic of investigation, and there have been many studies in this regard. Various Heat manufacturing techniques can be applied to heat sinks like machining casting or forging. If the heat sink features are too small or the geometry is too complex, this might become problematic for conventional manufacturing techniques. Here, 3D printing techniques could integrate these complex geometries into the fin design.

Tseng et al. did an experimental study to investigate the performance of water-cooled heat sinks. They explored different sinks structures for the study. The test samples are made from laser-based metal additive manufacturing (AM) process using a Titanium alloy for fabrication. Their test results showed that the proposed porous structure had a much lower pressure drop than the conventional metal foam. The proposed homogeneous porous structure has an enhanced performance compared to the reference sample [23]. Collins et al. investigated the application of the direct metal laser sintering (DMLS) additive manufacturing process to manufacture microchannels heat sinks using an aluminum alloy (AlSi10Mg). They demonstrated the heat sink's thermal and hydraulic performance by considering water as the working fluid. The knowledge acquired from the study was planned to be applied further, exploring the design of the heat sink utilizing various geometric complexities with additive manufacturing [24].

It is worth mentioning that additive manufacturing is a more feasible technique for fabricating complex parts than typical manufacturing [25]. Additive manufacturing techniques, particularly the Selective Laser Melting (SLM) technique, could produce complex features of heat sinks [26,27]. An experimental study performed by Lei et al. was on heat sink manufacturing optimization by applying Topology optimization. In the study, 3d printed patterns were produced using stereolithography and investment casting for

Table 1
Heat sink parameters.

Heat sink	Symbol	Slot-width, w (mm)	Slot-depth, d (mm)	Number of slots
Plain-fin heat sink	P	–	–	–
Slotted-fins heat sink	S1	1	17	1
	S2	1	34	1
	S3	2	34	1
	S4	3	34	1
	S5	1	34	2
	S6	2	34	2
	S7	3	34	2
	S8	1	34	3
	S9	2	34	3
	S10	3	34	3

manufacturing metal heat sinks designed earlier. They demonstrated that 3d printed patterns and investment casting are viable, practical, and precise techniques to produce metal heat sinks and other structurally optimized parts. To validate the results, simulations were performed to investigate further factors like temperature distribution, fluid flow patterns, and heat transfer for better performance under the given design conditions [28]. The growth of the boundary layer along the length of the plate fins can be interrupted by making slots in the fins. This study optimizes the slot size to achieve low temperature and pressure drop in an array of plate fins under turbulent airflows.

2. Numerical analysis

2.1. Problem description

The growth of the viscous boundary layer along the length of the plain fins is interrupted by making slots in the fins. Multiple parameters of the slots are varied to come up with the slotted fins with a higher heat transfer rate and lower temperatures. The geometric parameters varied for this research are the width and depth of the slot. Also, the number of slots is varied from 1 to 3. The slotted fins considered for the current study are described in Table 1.

Each heat sink (plain/slotted) consists of three identical fins on a 5 mm thick base. Each fin is 4 mm thick, and the edge-to-edge distance between two consecutive fins is 6 mm, as shown in Fig. 1(a). The width and depth of the slot are indicated with w and d, respectively, also shown in Fig. 1(b). The material of all heat sinks is Aluminum alloy with thermal conductivity of 239 W/m²K. The CAD models of the plain and slotted fins can be seen in Fig. 1.

2.2. Conjugate heat transfer model

When heat energy is applied to the heat sink base, it travels through the fins by conduction. Meanwhile, air flowing over the fins takes the heat away from the fins via convection. This phenomenon refers to the conjugate heat transfer.

In the current study, air enters the domain at 24 °C. The Re number of the flow is calculated using Eq (1). The numerical analysis was performed at the airspeeds (U_{air}) of 2, 4 & 6 m/s, resulting in Re numbers 13178, 26356, and 39534, respectively. Air density (ρ_{air}) and dynamic viscosity (μ_{air}) were considered at 24 °C and uniform throughout the analysis.

The hydraulic diameter (D_h) was calculated by using the cross-sectional area (A_c) and the perimeter (P) of the experimental test section, as shown in Eq (2).

$$Re = \frac{\rho_{air} U_{air} D_h}{\mu_{air}} \quad (1)$$

$$D_h = \frac{4A_c}{P} \quad (2)$$

Fourier's law of conduction (Eq (3)) was applied for the solution of heat conduction through the aluminum heat sinks.

$$\nabla \cdot (k_s \nabla T_s) = 0 \quad (3)$$

where.

k_s - the coefficient of conduction for aluminum, i.e., 239 W/m.K.

T_s - aluminum heat sink temperature

The Reynolds averaged Navier Stokes (RANS) (Eqs. (4) and (5)) were applied to the turbulent flow through the heat sink model. The time-averaging of the momentum, energy, continuity, and decay variables from mean & variable components were utilized to attain these equations. More studies were also performed using the RANS equations by other researchers like Zhou [29] and Leung [30].

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho = 0 \quad (4)$$

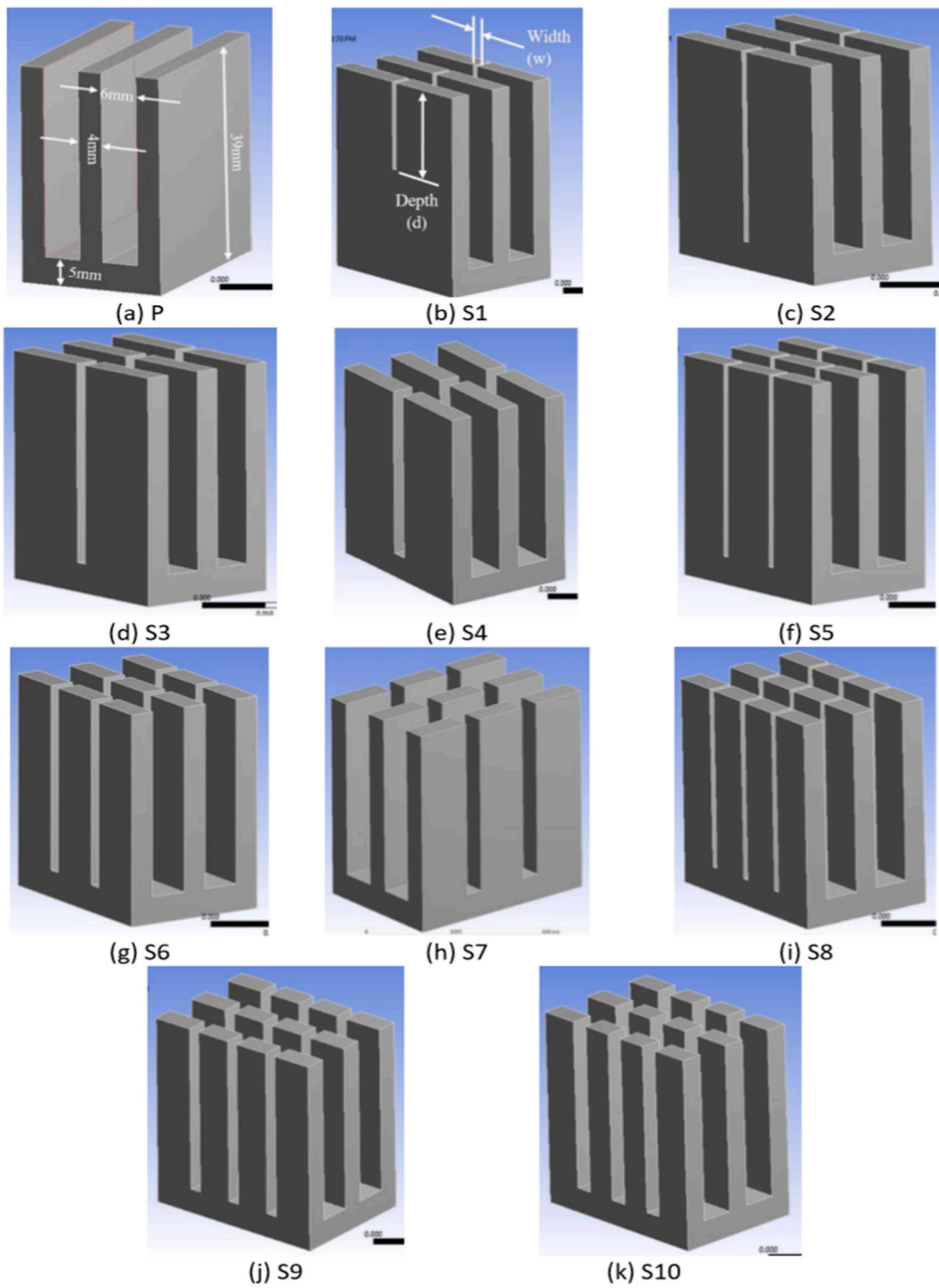


Fig. 1. CAD models of heat sinks.

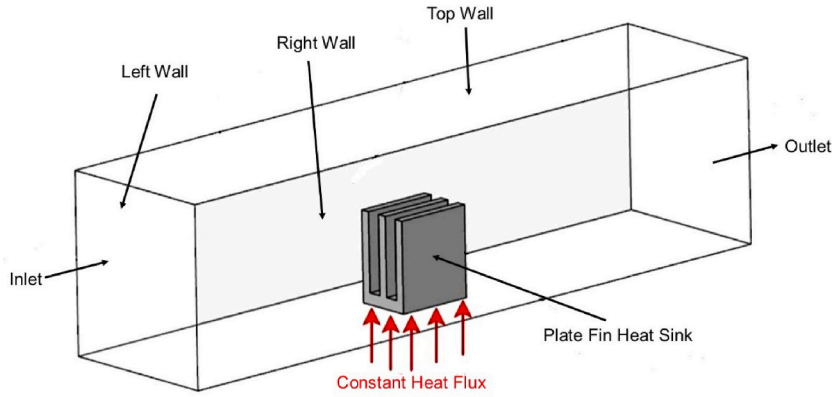


Fig. 2. Flow domain schematic for the CFD analysis.

$$\frac{\partial(\rho)}{\partial t} + \nabla \cdot (\rho) = \nabla \cdot (\bar{\sigma} - \rho) \quad (5)$$

where $\bar{\sigma} = -p\bar{I} + \mu(\nabla + |\nabla|^T)$ and $-\rho = \mu_t(\nabla + |\nabla|^T) - \frac{2}{3}(\rho k\bar{I})$ are the Newtonian and Reynolds stress tensors. p is the pressure density, and ρ, μ is the viscosity of air.

$\bar{U}$ and $\bar{U}'$ are standard turbulent fluctuation velocity vectors and $\bar{I}$ is a unit tensor.

The temperature area in the fluid (T_f) was computed by the RANS equation and the energy Eq. (7). RANS equations were utilized to determine the temperature field in the fluid (T_f) with the energy Eq. (7).

$$\rho \frac{\partial C_p T_f}{\partial t} + \rho \bar{U} \cdot \nabla (C_p T_f) = \nabla \cdot \left(k + \frac{C_p \mu_t}{Pr_t} \right) \nabla T_f + \dot{Q} \quad (6)$$

$\dot{Q}$ is the heat source (watts),
 C_p the specific air heat capacity, and
 Pr is the Prandtl number of air.

The subscript t shows the turbulent counterparts.

For simulating the thermal airflow through the heat sink, the k- ω Shear Stress Transport (SST) model (Eqs. (7)–(13)) is applied. Previously, the model was utilized by other researchers like Zhou [29] and Leung [30] to calculate separated flow behavior. It merges the k- ω factor to accurately calculate the turbulent thermal airflow through the aluminum heat sink. The model utilized the automatic wall function, which ignored radiation heat transfer. The air density and viscosity values should be steady and equal to the inlet at 24 °C.

$$\frac{\partial(\rho k)}{\partial t} + \bar{U} \cdot \nabla (\rho k) = \tilde{P}_k - \beta^* \rho k \omega + \nabla \cdot [(\mu + \sigma_k \mu_t) \nabla k] \quad (7)$$

$$\frac{\partial(\rho \omega)}{\partial t} + \bar{U} \cdot \nabla (\rho \omega) = \alpha \rho S^2 - \beta \rho \omega^2 + \nabla \cdot [(\mu + \sigma_\omega \mu_t) \nabla \omega] + 2(1 - F_1) \rho \sigma_{\omega_2} \frac{1}{\omega} \nabla k \cdot \nabla \omega \quad (8)$$

where F_1 is the blending function.

$$F_1 = \tanh \left(\left\{ \min \left[\max \left(\frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right), \frac{4\rho \sigma_{\omega_2} k}{CD_{kw} y^2} \right] \right\}^4 \right) \quad (9)$$

in which,

$$CD_{kw} = \max \left(2\rho \sigma_{\omega_2} \frac{1}{\omega} \nabla k \cdot \nabla \omega, 10^{-10} \right) \quad (10)$$

The turbulent eddy viscosity is defined by,

$$\nu_t = \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (11)$$

where S and F_2 show constant strain rate and a second blending function, respectively.

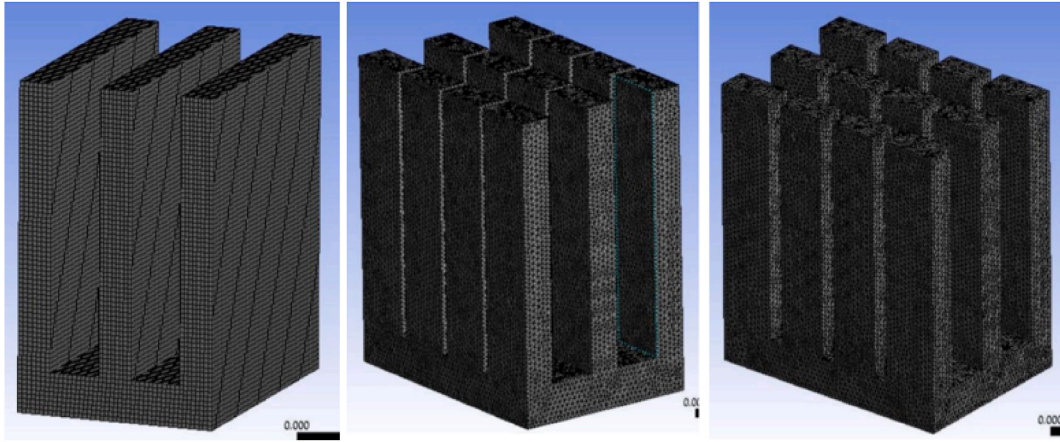


Fig. 2a. The hexahedral mesh of heat sink P and tetrahedral mesh of heatsinks S8 and S9.

Table 2

Mesh independence test.

Heatsink	Number of cells	T_{base} (°C)	ΔP (Pa)
P	159861	49.8	0.7
	232374	50.5	1.0
	275276	50.6	1.1
S8	816374	48.8	0.8
	874685	49.4	1.1
	981793	49.5	1.2
S9	808487	46.9	1.0
	989642	47.8	1.3
	991835	47.9	1.3

$$F_2 = \tanh \left(\left[\max \left(2 \frac{\sqrt{k}}{\beta^* \omega y}, \frac{500\nu}{y^2 \omega} \right) \right]^2 \right) \quad (12)$$

In the stagnation regions, a production limiter (P_k) is applied to restrict the progression of turbulence.

$$P_k = \mu_t \frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \rightarrow \tilde{P}_k = \min(P_k, 10\beta^* \rho k \omega) \quad (13)$$

The constants in this SST model are $\beta^* = 0.09$, $\alpha_1 = \frac{5}{9}$, $\beta_1 = \frac{3}{40}$, $\sigma_{k1} = 0.85$, $\sigma_{\omega_1} = 0.5$, $\alpha_2 = 0.44$, $\beta_2 = 0.0828$, $\sigma_{k2} = 1$ and $\sigma_{\omega_2} = 0.856$.

2.3. Computational domain and boundary conditions

The reverse airflow in the computational domain is avoided by significantly having the inlet and outlet from the heatsink. The heatsink was placed in the middle of the environment, as shown in Fig. 2.

For the CFD analysis, boundary conditions are as follows:

1. The air temperature at the inlet (T_{in}) is 24 °C.
2. Air velocity (U_{air}) ranges from 2 m/s to 6 m/s.
3. Air pressure at the outlet, $P = P_{\text{gauge}} = 0$ Pa.
4. No-slip condition at the top, right and left walls.
5. $\frac{dT}{dx} = 0$, $\frac{dT}{dy} = 0$ and $\frac{dT}{dz} = 0$
6. A uniform heat flux of $Q = 8578.4 \text{ W/m}^2$ at the heat sink base.
7. Considering a no-slip condition at the air and heatsink interface, the heat flux is conserved. $k_f \frac{dT_f}{dn} = k_s \frac{dT_s}{dn}$

2.4. Solution methods and convergence criteria

The SIMPLE technique or the Semi Implicit Method for Pressure Linked Equations is applied for solving the continuity equations. ANSYS FLUENT code was used for the coupled energy and momentum equations. The resolution of the energy equations was considered converged when the sum of the residuals is less than 10^{-6} , utilizing the second-order upwinding for the solution. The solution of the continuity and momentum equations was considered combined when their residuals sum is less than 10^{-4} .

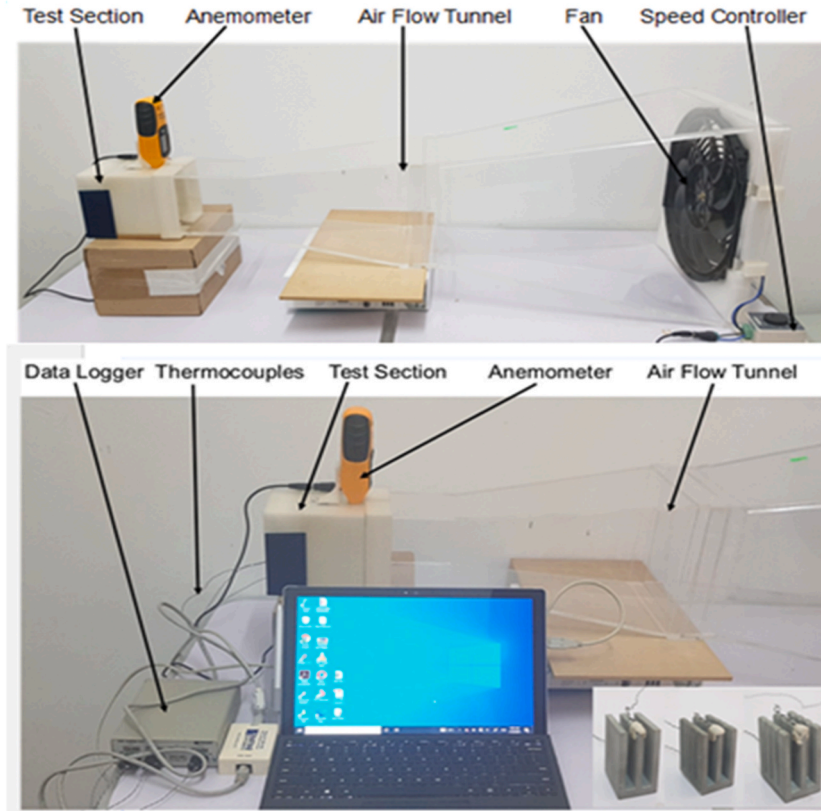


Fig. 3. Wind tunnel, test section, data acquisition, and cast Aluminium Fins with thermocouples.

2.5. Grid independence test

The heat sinks are divided into small and discrete elements until mesh independence is achieved. The features for the plain-fins heat sink are hexahedral, while for all other slotted-fins heat sinks, the cell shape is tetrahedral. The heatsink P has hexahedral mesh, while the tetrahedral mesh is used for the rest of the heatsinks, including S8 and S9, as shown in Fig. 2.1 below.

The heatsink P, S8 and S9 achieve grid independence after being split into 275276, 981793, and 991835 discrete cells, as shown in Table 2. Mesh independence is considered when the difference between two consecutive iterations of T_{base} and ΔP is less than 0.5%.

2.6. Experimental analysis

The aluminum heatsinks were fabricated using Digital Metal Casting technology. For manufacturing the casting patterns of the heatsinks, the Fused Filament Fabrication (FFF) process was applied using the Ultimaker-3 3D printer to produce the patterns of the heat sinks P, S8, and S9. These 3D printed patterns were used in the investment casting process to create aluminum heatsinks for the experiments, as shown in Fig. 3.

After developing heatsinks, thermocouples were embedded at the base and the tip of the heatsinks using thermally conductive epoxy, as shown in Fig. 3. For experimental analysis, calibrated J-type thermocouples measure the temperature of the heatsink and the air around it. The heatsinks were placed inside the test section (cross-section inlet area 100 mm × 100 mm), developed using the FFF technique, and housed all the cutouts for the heat sink and thermocouples. A wind tunnel was designed and fabricated with acrylic material to accommodate the test section and a variable-speed 12v D.C. fan. The actual experimental setup is shown in Fig. 3. Its power is controlled by varying the current passing through the fan, maintaining its speed ($P = VI$). An anemometer was used to measure the airspeed inside the test section at different fan powers. A hexagonal flow straightener was used at the air-inlet to homogenize the airflow inside the test section. A 7-W film-heater was placed at the base of the heat sinks. The temperature data is recorded using a data acquisition system and fed to the computer. The bottom and tip temperatures were recorded for the heat sinks P, S8, and S9, at airspeeds of 2, 4 & 6 m/s, as shown in Fig. 4. The heat sink base was always at a higher temperature than the fin tip. The base and tip temperatures start rising for each case when the heat sink is provided with heat flux and airflow, called the transient state. After a while, the temperatures become constant, called the steady-state temperature. For each case, the steady-state temperatures were different. The higher steady-state temperatures were observed for lower airspeeds and vice versa.

The temperatures of the heat sink and the air around it were used to calculate the convection coefficient (h) at different airspeeds for the heat sinks P, S8, and S9. The heat transfer coefficient was calculated using Eq (14). The heat transfer through radiation was ignored, and the primary mode of heat transfer is convection. Therefore, heat supplied to the heat sink (Q) equals the heat dissipated

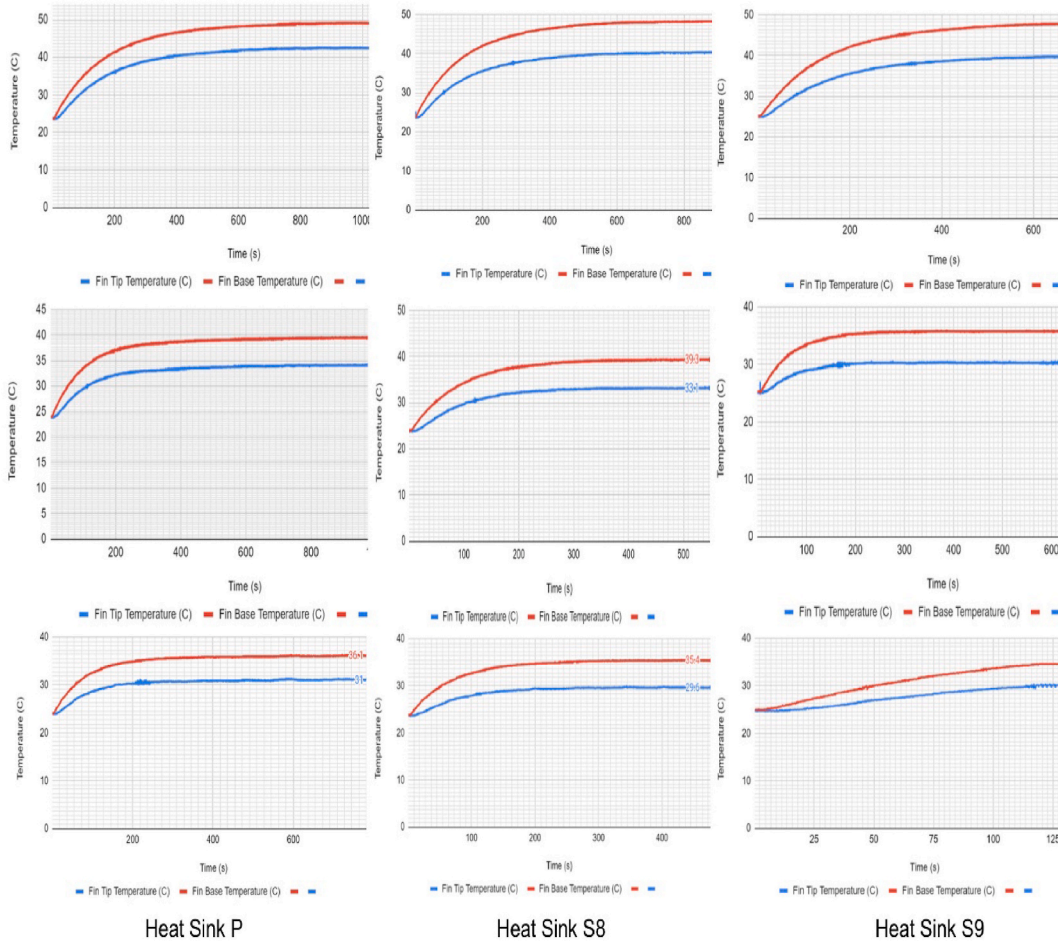


Fig. 4. Experimental temperatures of the heat sink P, S8 & S9 at airspeed = 2, 4 & 6 m/s.

through convection.

$$h = \frac{Q}{A_s(T_s - T_\infty)} \quad (14)$$

where.

T_s is the mean top surface temperature of the sink,

T_∞ is the mean temperature at the test section's inlet/outlet

A_s is the surface area of the heat sink responsible for convection heat transfer.

3. Results and discussion

3.1. Thermal performance

The CFD analysis of the plain and slotted heat sinks provides insights into the effect of different aspects of the slots on the heat sinks heat transfer. It was noticed that slots are beneficial for the enhancement of the heat transfer rate, shown by the relatively lower temperatures of the heat sinks S1, S2, S3, S4, S5, S6, S7, S8, S9, and S10 than those of heat sink P, as shown in Fig. 5. The increase in the depth of the heat sink slot decreases the base and tip temperatures of the heat sink, as shown in Fig. 5 (b) and (c). Increasing the width of the slot (w) shows exciting results, as increasing the width from 1 mm to 2 mm enhances the heat transfer, while increasing the width from 2 mm to 3 mm decreases the heat transfer rate. This might be because of optimal airflow in the 2 mm wide slots of the heat sinks S3, S6, and S9. The number of slots is also varied from 1 to 3 in various heat sinks, and results indicated that an increased number of slots enhances the heat sink heat transfer rate. Fig. 6 suggests that the heat sinks having three slots exhibit relatively lower base and tip temperatures for the same width and depth of slots.

In the temperature contours of all the heat sinks, it was observed that the windward end for the heat sink is at a slightly reduced temperature than the rest. As the air initially affects the heat sink, it is at its lowest temperature, providing the maximum temperature

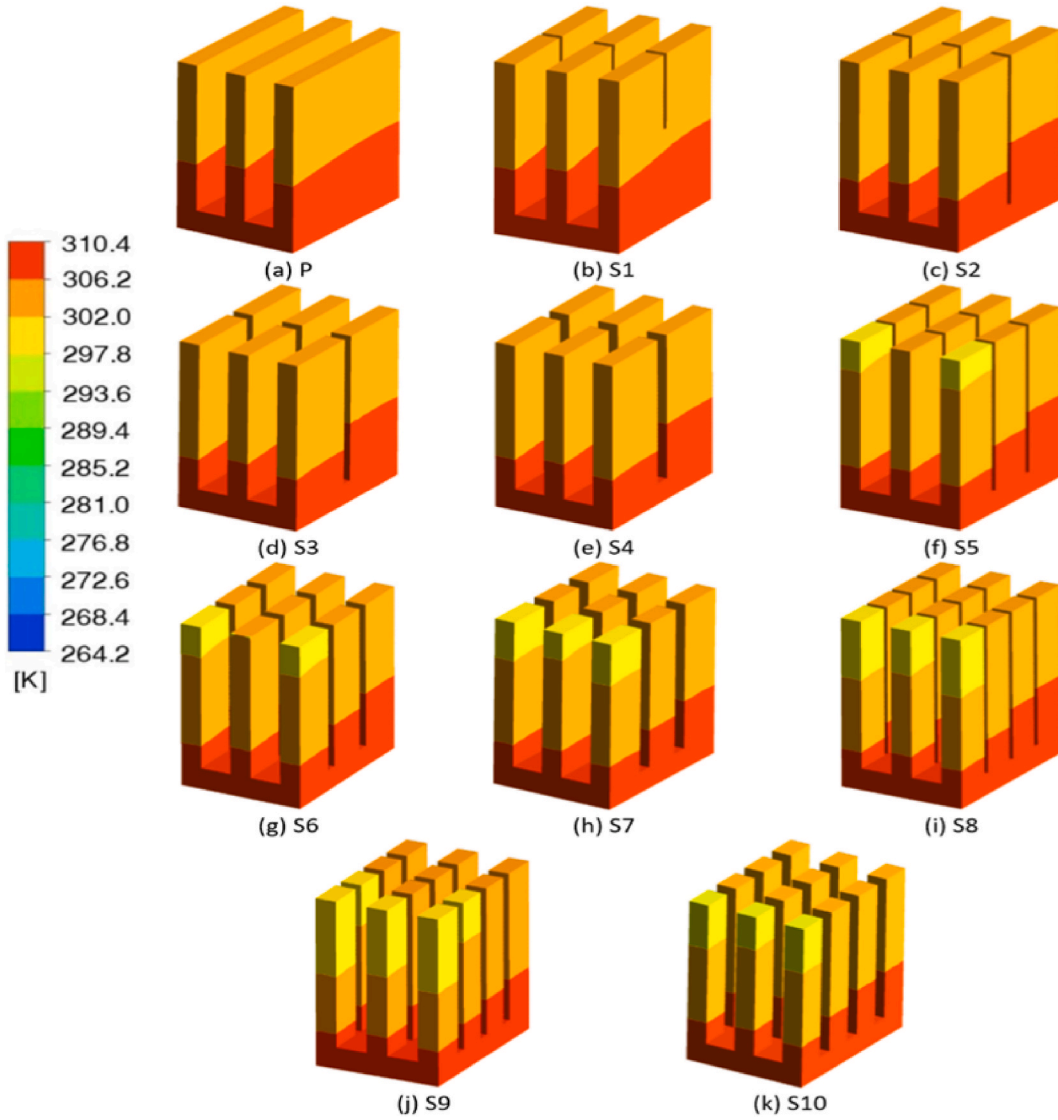


Fig. 5. Temperature distribution in plate-fin heat sinks at an airspeed of 6 m/s.

gradient between solid and the atmosphere. With air convection, some of the heat from the leading end spreads over the rest, providing a relatively lower temperature gradient and thus resulting in slightly elevated temperatures.

The heat sinks S8 and S9 result in the lowest possible temperatures at each wind speed. Both heatsinks have three full-depth slots. Fig. 6 shows that the heatsink S9 has the most inferior tip and base temperatures at 2, 4, and 6 m/s, respectively, followed closely by the S8. Fig. 7 indicates the Nu number of the heat sinks P, S8, and S9 against the Re number of the flow at 2, 4 & 6 m/s. The results indicated that the increase in the Re number also enhances the Nu number of every heat sink. The heat sinks S8 and S9 have a higher Nu number than the heat sink P at each Re number. The heat sink S9 results in the highest Nu number at each Re number, higher than the Nu number of P by 28.4% on average. The heat sink S8 has an average 8.8% higher Nu number than the heat sink P. Thus, for the best thermal performance, the heatsink S9 is the best choice among all the heat sinks.

3.2. Experimental uncertainty analysis

Uncertainty in the convection coefficient (σ_h) is the function of the uncertainties in the surface temperature (σ_{T_s}) and ambient temperature (σ_{T_∞}) measurements.

$$\sigma_h = f(\sigma_{T_s}, \sigma_{T_\infty}) \sigma_h = \pm \sqrt{\left(\frac{\partial h}{\partial T_s} \sigma_{T_s} \right)^2 + \left(\frac{\partial h}{\partial T_\infty} \sigma_{T_\infty} \right)^2}$$

The maximum uncertainties in the measurement of h for heat sink P, S8, and S9 are ± 0.81 , ± 0.78 , and ± 0.91 , averagely at

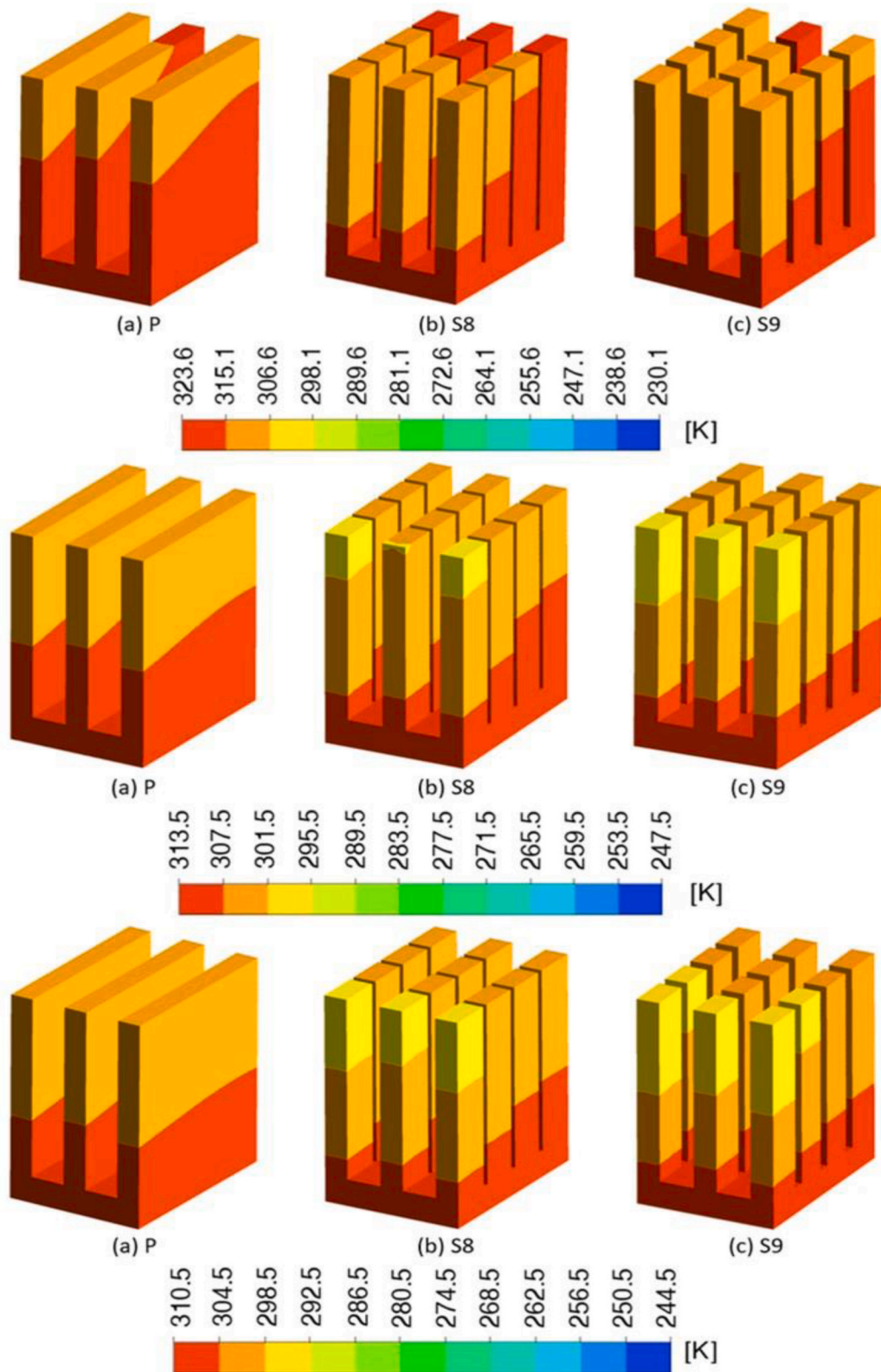


Fig. 6. Temperature distribution in plate-fin heat sinks at an airspeed of 2, 4 & 6 m/s

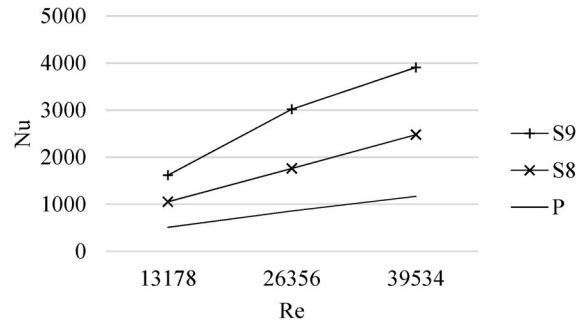


Fig. 7. Nu number of heatsinks against Re number.

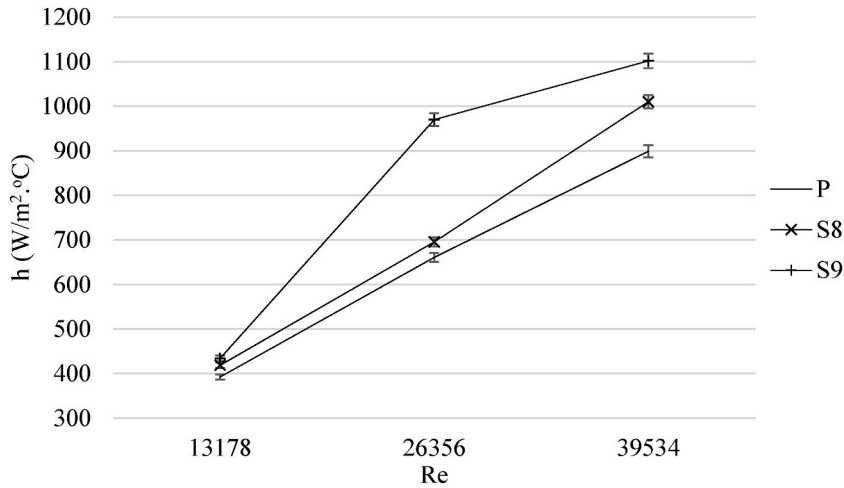


Fig. 7a. Convection heat transfer of heatsinks at different Re numbers

different airspeeds.

The relative uncertainty of the observation is ± 0.081 , ± 0.004 , and ± 0.0045 for heat sink P, S8, and S9, respectively, which is calculated using the relation below.

$$U_h = \frac{\pm \sqrt{\left(\frac{\partial h}{\partial T_s} \sigma_{T_s}\right)^2 + \left(\frac{\partial h}{\partial T_\infty} \sigma_{T_\infty}\right)^2}}{h}$$

3.3. Experimental repeatability analysis

The experiments are repeated five times, and the maximum percentage difference between calculated h values is 1.5%, as depicted in Fig. 7.1.

3.4. Hydraulic performance

The hydraulic analysis of the thermally optimized heat sinks was performed, and the pressure drop through the heat sinks was calculated. The air pressure at the heatsink's inlet is high and lowers down as the wind flows over the heat sink. The pressure drop through the heat sink holds significance, as higher fan power is required to push the air through a heat sink which causes a higher pressure drop. The pressure drop (Pa) is calculated at each Re number for the heat sinks P, S8, and S9, and the pressure contours around the heat sinks are shown in Fig. 8. The heat sinks act as hindrances in the airflow path, decreasing the air pressure. The higher the pressure drop is observed, the higher the windspeed for each heat sink. The pressure drop is the lowest across the heatsink P and the highest across S9 for each Re number, with the S8 value lying in between. Fig. 9 shows that the pressure drop across the heat sink S8 and S9 is 2.83% and 9.31% higher than the pressure drop across the heatsink P. The slotted fins (S8 and S9) dropped pressure caused by enhanced hindrance in the airflow path due to the presence of slots. However, where slots enhanced the heat transfer rate, they also introduced a pressure drop in the airflow path.

3.5. Validation of numerical model

The convection coefficient attained from the experimental temperature values is used to calculate the temperature field in the heat

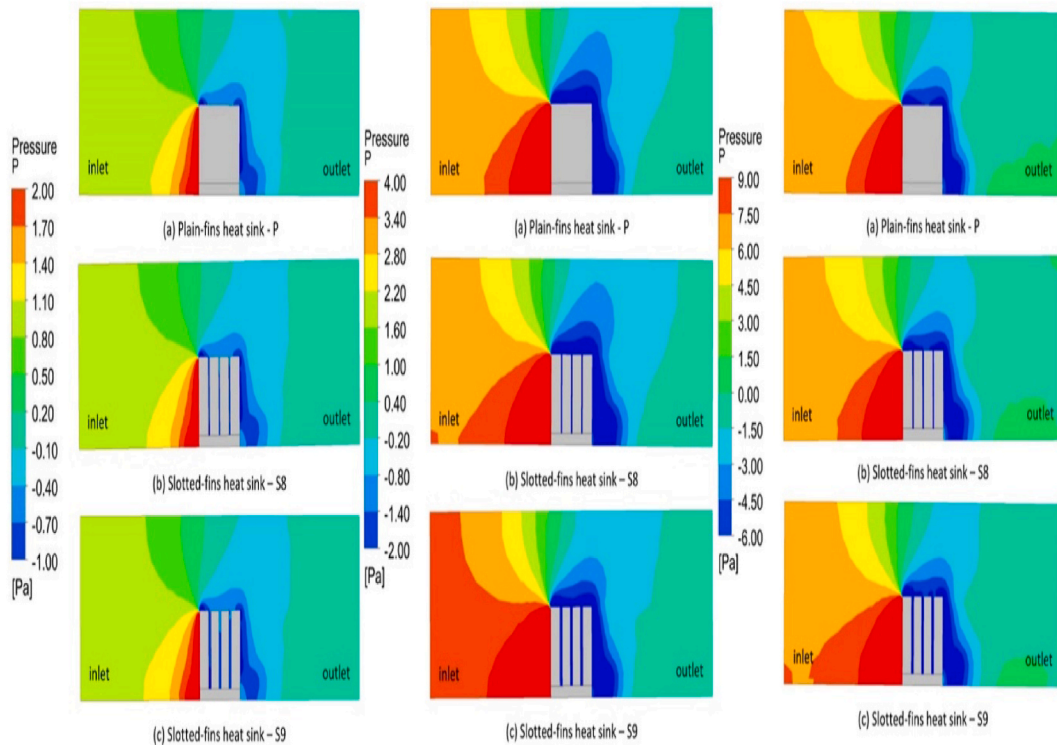


Fig. 8. Pressure distribution across plate-fin heat sinks at an airspeed of 2,4 & 6 m/s

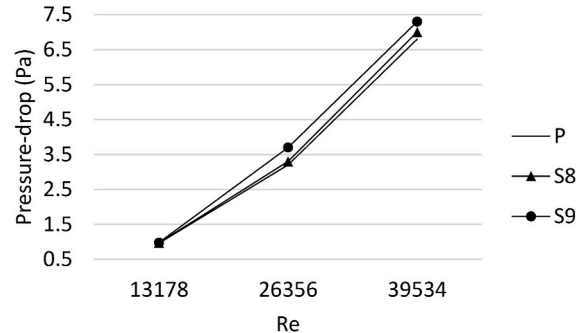


Fig. 9. Pressure-drop across different heat sinks at different Re number.

sinks using steady-state thermal analysis in ANSYS. The heat sink material is aluminum with thermal conductivity of $239 \text{ W/m}^\circ\text{K}$. The heat applied to the heat sink base is 7 W , and the ambient temperature for analysis is assumed to be 24°C . The temperature distribution in the heat sinks P, S8, and S9 at different airspeeds is shown in Fig. 8. For comparison of the simulated and experimental results, the base temperatures of the heat sinks were plotted against different airspeeds in Fig. 10. The maximum difference between the observed and simulated values is 2.6%, 2.9%, and 3% for the heat sinks P, S8, and S9.

4. Conclusion

To solve the overheating problem in the current study, the geometry of heat sinks was modified for either enhanced surface area or interrupted viscous boundary layer of the cooling fluid. In the present research, slots were fabricated for the plate fins to maximize the availability of the surface area for heat transfer during forced convection by air. The geometry of the slots (depth and width) was optimized for this purpose. Additionally, slots varied from 1 to 3 for efficient thermal performance. The CFD analysis, validated with the experimental analysis, indicated that most of the variations of slotted plate fins result in lower temperatures than the plain fins. The plate-fin heatsink with the best thermal performance was S9 (having three 34 mm deep and 2 mm wide slots). It was concluded that, for all airspeeds, the base temperature of the heatsink S9 is 5.6% lower than the heatsink P, while the convection coefficient of S9 is 28.4% higher than that of the heatsink P, on average. However, the formation of slots reduces the air pressure across the heat sink, and S9

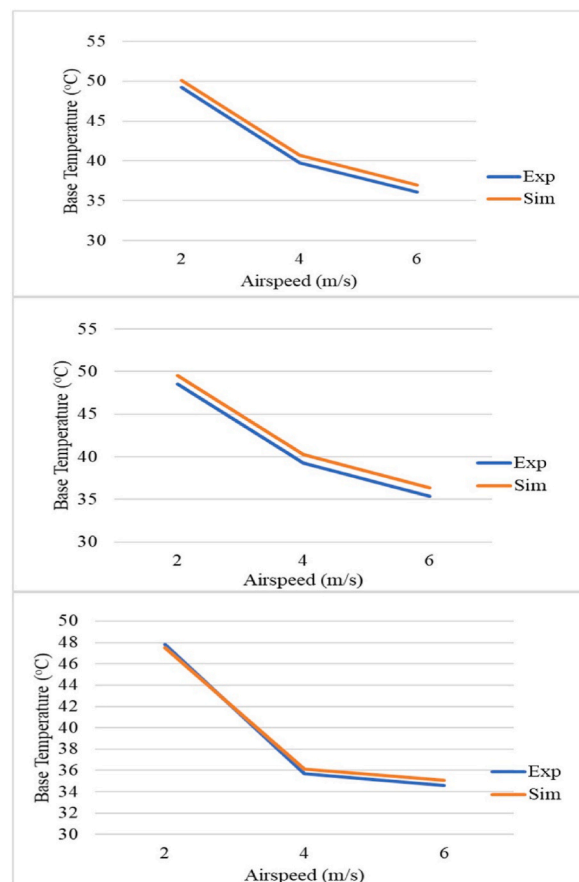


Fig. 10. Base temperature of the heat sink P, S8 & S9 at different airspeeds.

causes a pressure drop of 9.3% higher than that of the heatsink P.

Author contributions

Khurram Altaf: Conceptualization, Resources, Funding acquisition, Supervision, Adeel Tariq, Syed Waqar Ahmad: Writing, Validation, Original draft preparation, Ghulam Hussain: Resources, Visualization, T.A.H. Ratlamwala: Writing, Review & Editing, Hafiz Muhammad Ali: Data Curation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The Authors confirmed that there is no CONFLICT OF INTEREST for this manuscript.

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